

CWA SCIENCE CLASSES

Concept With Abhishek | Class 12 — Physics

Chapter 1: Electric Charges and Fields

1. Electric Charge: Fundamental property of matter; two types — positive and negative. Like charges repel, unlike charges attract.
2. Quantisation of Charge: Charge on any object is an integral multiple of e (charge of electron), $q = ne$.
3. Coulomb's Law: $F = k \frac{q_1 q_2}{r^2}$, where $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$.
4. Electric Field: Region around a charge where another charge experiences a force. $E = F/q_0 = kQ/r^2$.
5. Electric Field Lines: Start from positive charge and end at negative charge; never intersect each other.
6. Electric Dipole: Pair of equal and opposite charges separated by a small distance; dipole moment $p = q \times 2a$.
7. Gauss's Law: Total electric flux through a closed surface = $q_{\text{enclosed}} / \epsilon_0$. Used to find field due to symmetric charge distributions easily.